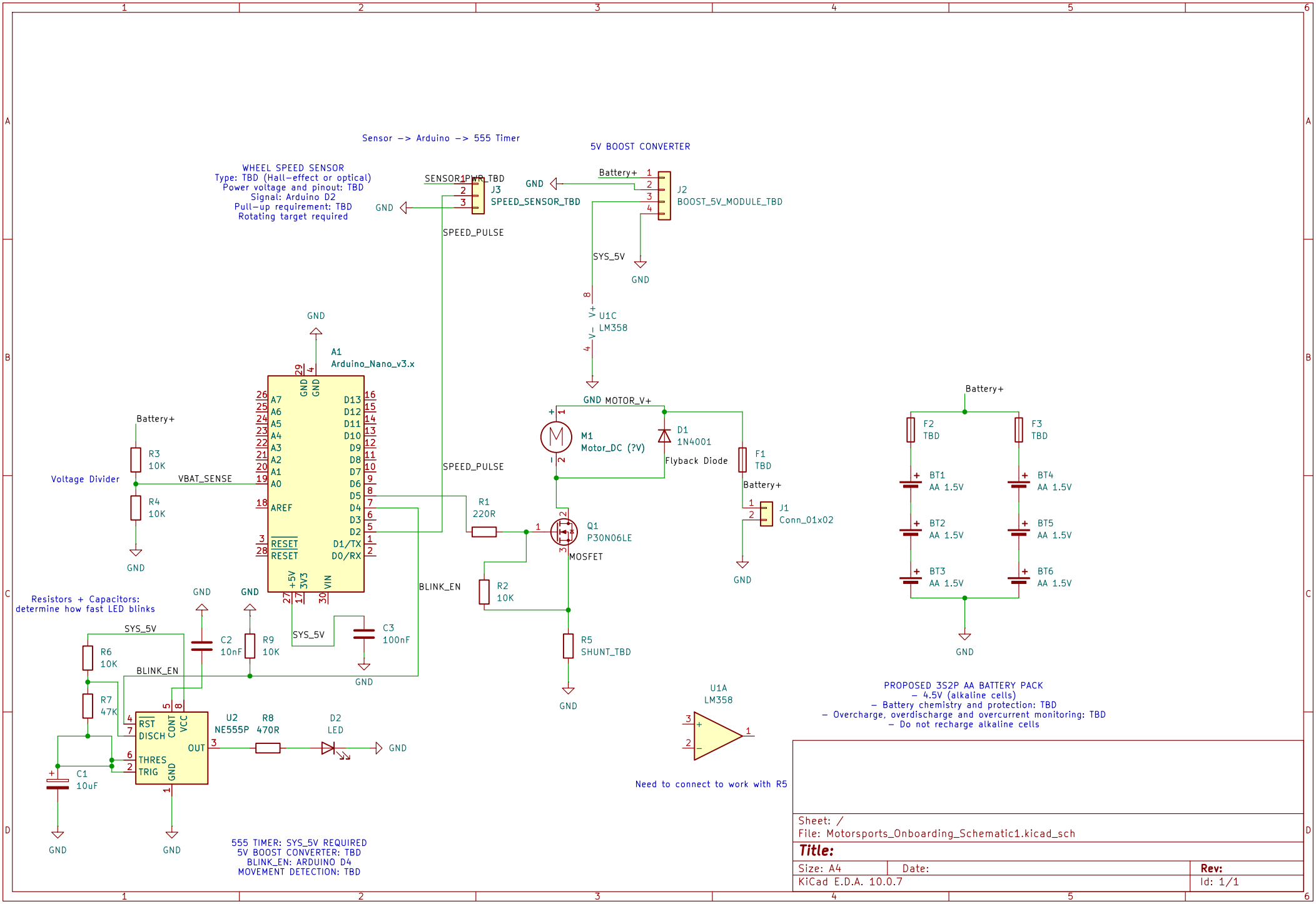

Sensor –> Arduino –> 555 Timer
5V BOOST CONVERTER
WHEEL SPEED SENSOR
Type: TBD (Hall–effect or optical)
Power voltage and pinout: TBD
Signal: Arduino D2
Pull–up requirement: TBD
Rotating target required
SENSOR1PWR_TBD
GND
J3
SPEED_SENSOR_TBD
SPEED_PULSE
Battery+
J2
BOOST_5V_MODULE_TBD
SYS_5V
U1C
LM358
A1
Arduino_Nano_v3.x
Voltage Divider
R3
10K
R4
10K
VBAT_SENSE
R1
220R
M1
Motor_DC (?V)
MOTOR_V+
D1
1N4001
Flyback Diode
F1
TBD
J1
Conn_01x02
Q1
P30N06LE
MOSFET
R2
10K
R5
SHUNT_TBD
BLINK_EN
Resistors + Capacitors:
determine how fast LED blinks
R6
10K
R7
47K
C2
10nF
R9
10K
C3
100nF
C1
10uF
U2
NE555P
R8
470R
D2
LED
555 TIMER: SYS_5V REQUIRED
5V BOOST CONVERTER: TBD
BLINK_EN: ARDUINO D4
MOVEMENT DETECTION: TBD
U1A
LM358
Need to connect to work with R5
F2
TBD
F3
TBD
BT1
AA 1.5V
BT2
AA 1.5V
BT3
AA 1.5V
BT4
AA 1.5V
BT5
AA 1.5V
BT6
AA 1.5V
PROPOSED 3S2P AA BATTERY PACK
– 4.5V (alkaline cells)
– Battery chemistry and protection: TBD
– Overcharge, overdischarge and overcurrent monitoring: TBD
– Do not recharge alkaline cells
Sheet: /
File: Motorsports_Onboarding_Schematic1.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 1/1